

220308 - Aircraft Propulsion Systems

Project: Real engine for civil applications

You are an aerospace engineer working in a new company whose name is yet to be defined. The company wants to develop new **propfan engines** that are competitive with the existing turbofan engines. As a first step, the company wants to get a fundamental **understanding of the current turbofan engines which later could be converted**. As a propulsion engineer, you have been assigned within a team of **4 to 6 people** the task of evaluating one of these turbofans.

The project leader, in an attempt to unify the outputs from all the research teams, has broken the tasks in the following parts:

1. Selection of the engine and required parameters. Search for a real turbofan engine (check supplementary material) for the analysis. Remember, its flight Mach number at cruise must be subsonic and you cannot choose the CFM56 (this has already been analyzed). Select all the relevant parameters of the engine for the design point:

- i. Cruise condition: h_0 and M_0 (assume standard atmosphere if needed). *Cannot be sea-level static.*
- ii. **Engine mass flow**. For that use the design point according to figure **Figure 1**.
- iii. Bypass ratio.
- iv. Engine parameters and efficiencies. You may assume current technology levels in case some of them cannot be found.
 - For the HPC, consider the design point given in Figure 1 and obtain its pressure ratio and efficiency from there.
 - For the LPC, use the OPR to obtain the pressure ratio.
- v. Fuel and T_{t4} .

2. On-design analysis and selection of the design point. Solve the thermodynamic cycle at the selected design point and obtain its performance parameters, i.e., F , $TSFC$, η_T , η_P and η_O . Then, produce heatmaps by changing the pressure ratios π_{LPC} and π_{HPC} . For that consider two scenarios:

- i. $\dot{m}_{core}$ and η_{HPC} are constant and equal to the design point.
- ii. $\dot{m}_{core}$ and η_{HPC} change according to the operating line on Figure 1.

Change π_{HPC} only within the limits of Figure 1. With respect to π_{LPC} , change at your discretion, within reasonable limits of OPR. Also remember to indicate the design point in the heatmaps.

3. Turbomachinery design. This part is divided between the HPC and HPT design.

- i. Design HPC assuming a repeating-row, repeating-stage cascade:
 - select a set of suitable parameters (M_1 , number of stages, etc.) from Figure 2;
 - draw and compute the velocity triangles;
 - solve the thermodynamic cycle.
- ii. Design HPT assuming one stage, constant axial velocity and zero swirl angle:
 - select a set of suitable parameters (R_t , Ψ , Φ , etc.) from Figure 3;
 - draw and compute the velocity triangles;
 - solve the thermodynamic cycle.

4. Off-design analysis. It is desired to understand how the engine performs at sea-level static conditions. For that solve the thermodynamic cycle assuming the required parameters from the design point selected. Obtain its performance parameters, i.e., F , $TSFC$, η_T , η_P and η_O .

Reporting. The project leader requests a report in PDF format as well as the tools used to perform the computations in a separate ZIP file. In your report remember to include the following items:

- Introduction presenting the selected turbofan with a summary table containing all the selected parameters and the reference from where they have been extracted. It is recommended to also include an engine diagram.
- A walkthrough of the computations. *No need to explain the theory, imagine this is a report for the project leader who has an understanding on propulsion..*
- A summary and evaluation of your results. Do include a summary table of your results for the requested points as well as the necessary sketches and drawings. Highlight all the relevant points.
- A conclusion giving an overall evaluation of the engine.

What follows on this document are the supplementary material referenced in this document.

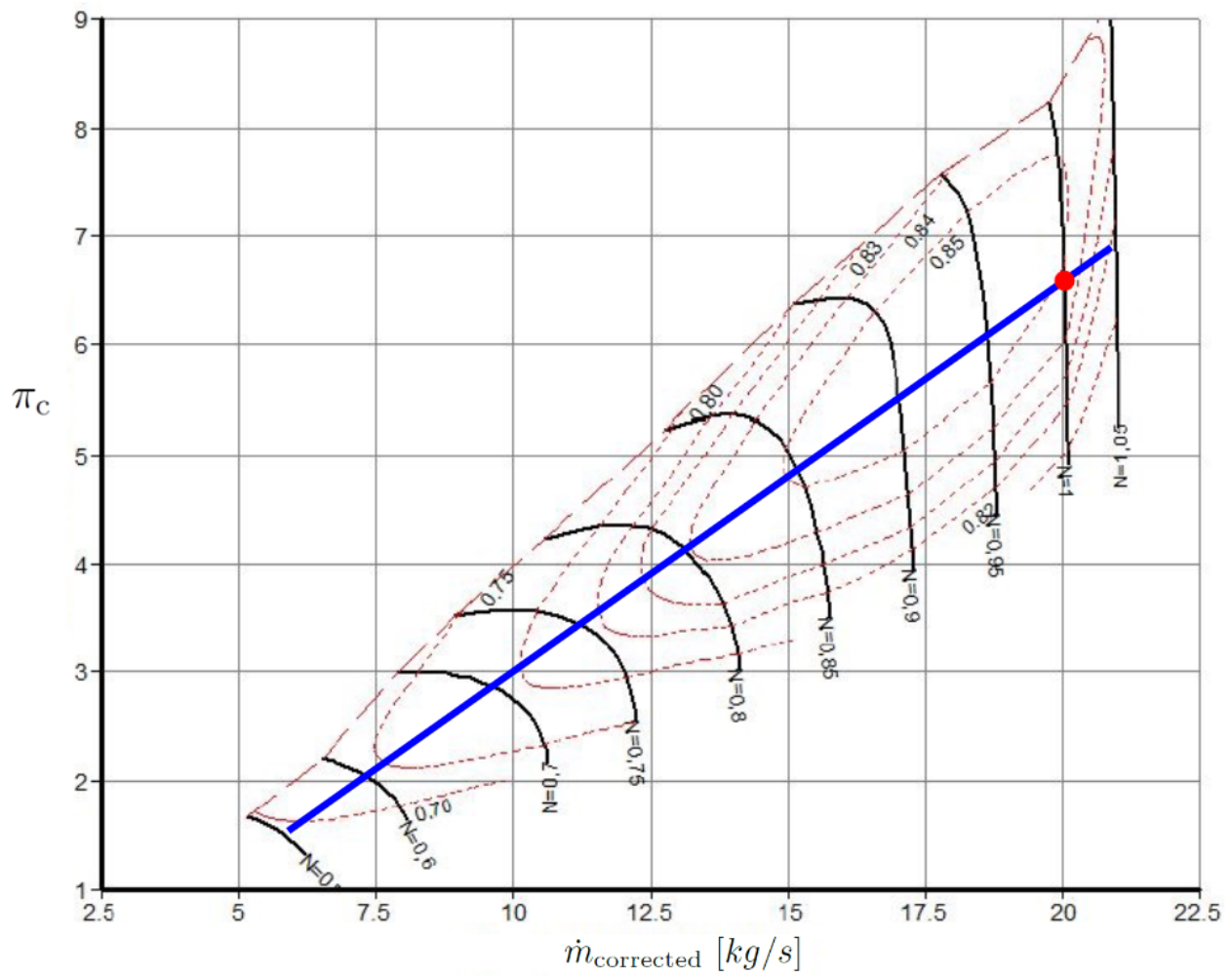


Figure 1: High pressure compressor (HPC) map. The blue line indicates the operating line while the red dot denotes the design point.

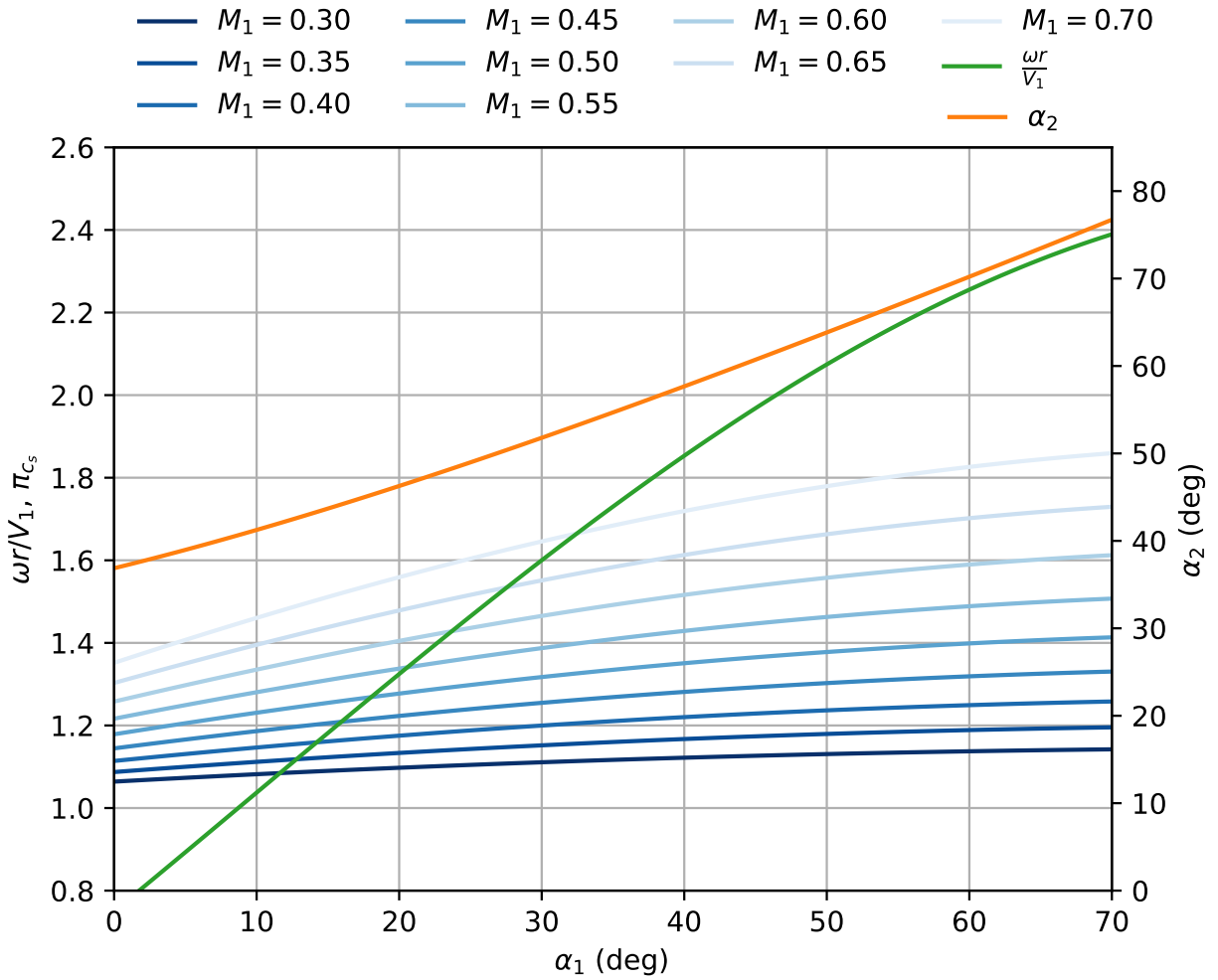


Figure 2: High pressure compressor (HPC) design for repeating-row, repeating-stage.

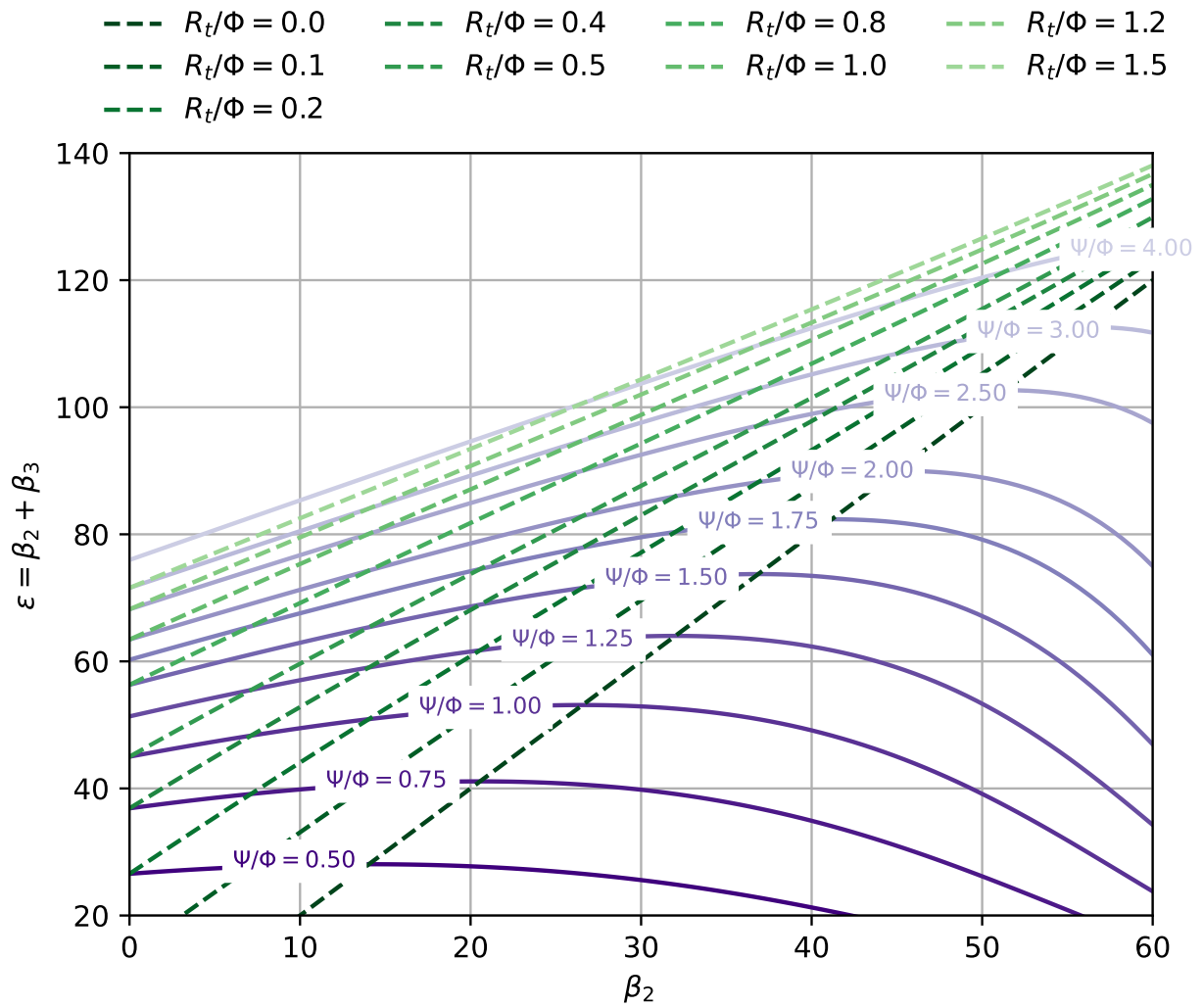


Figure 3: High pressure turbine (HPT) design for constant axial speed.